Agentic Portal: A Unified AI Agent Gateway & Orchestration MVP

Agenda

1) **Project Requirements Overviews**

2) **Project Scope**

3) **Core Architecture: Intelligent Routing & MCP Aggregation Gateway**

4) **Milestones & Acceptance Criteria**

5) **Future Evolution Plan**

Project Requirements Overviews (4 perspectives)

1. Unified Interaction & Experience

- **Single-Entry Portal Architecture**

Establish a centralized Web portal to eliminate the friction of switching between multiple AI applications and chatbots.

- **Natural Language Intent Interaction**

Support fully intent-driven inputs, enabling users to invoke diverse AI capabilities without memorizing specific commands.

2. Intelligent Routing & Memory Engine

- **Dynamic Intent Routing**

Autonomously dispatch queries between MCP tools, vertical bots, and general LLMs based on real-time intent analysis.

- **Session Context Persistence (MVP Core)**

Maintain session-level memory to support pronoun resolution and automated parameter completion in multi-turn dialogues.

- **Automated Entity Extraction**

Identify and extract key entities from conversations to automatically populate tool execution schemas.

Project Requirements Overviews (4 perspectives)

3. Standardized Integration & Extensibility

- **Standardized MCP Integration**

Leverage Model Context Protocol (MCP) as the core standard to achieve "plug-and-play" tool integration via an aggregated gateway.

- **Heterogeneous System Compatibility**

Integrate existing vertical chatbots via standard APIs to protect legacy IT investments.

- **Metadata-Driven Configuration**

Enable agent onboarding via metadata updates, eliminating the need to modify the portal’s core codebase.

4. Security, Monitoring & Governance

- **Granular Access Control**

Implement Role-Based Access Control (RBAC) to provide agent-level auditing and protection for sensitive tools and data.

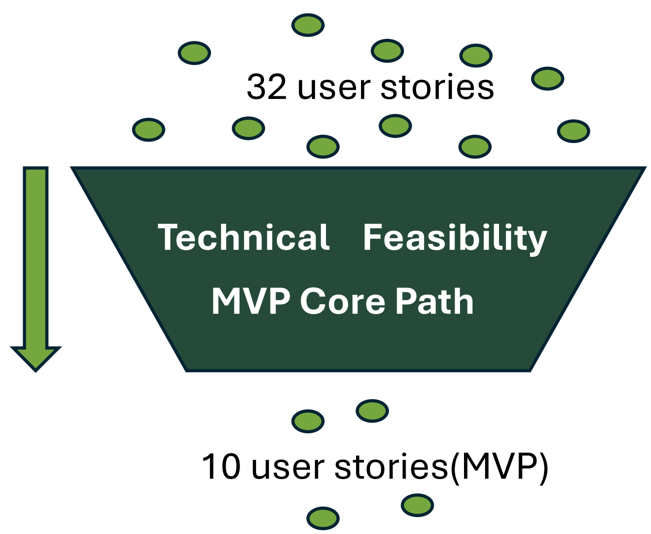
- **Full-Chain Execution Monitoring**

Track the entire lifecycle from intent routing to tool execution for performance monitoring and troubleshooting.

Project Scope

The core goal is to establish :

A **unified conversational entry point** that can identify user intent through **intelligent routing**, invoke different AI capabilities via the **MCP aggregation gateway**, or call upon third-party chatbots through **standard APIs**.



Core Architecture: Intelligent Routin Gateway

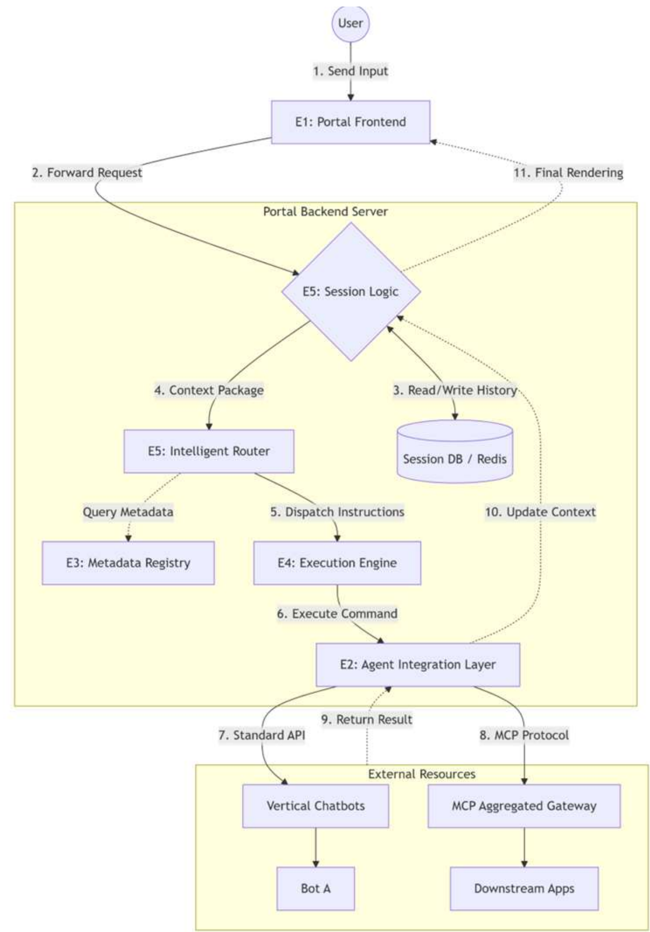
E1: Portal Frontend
Provides a unified conversational interface for natural language input

E2: Agent Integration Layer
Acts as a protocol bridge to adapt MCP and standard APIs, enabling seamless integration and metadata normalization for heterogeneous agents

E3: Agent Discovery & Overview
Maintains a centralized registry of agent semantic descriptions and parameter schemas to support dynamic capability discovery by the LLM.

E4: Agent Execution
Manages task lifecycles, including concurrent dispatching, timeout control, and error handling to ensure stable and efficient execution.

E5: Intelligent Routing & Session Logic
Analyzes user intent using LLMs and integrates session history for context-aware routing and automated fallback to general LLMs.



Milestones & Acceptance Criteria

M1: Foundation & Protocol Prototyping (Weeks 1-5)

Core Objective: Establish backend infrastructure and ensure connectivity with external tools.

1.1 Session Storage
AC: Implement Session-ID indexed storage in Redis/SQLite; ensure full message history is persistent and retrievable.

1.2 MCP Gateway Integration
AC: Establish stable connection with MCP Aggregated Gateway; successfully retrieve downstream tool metadata

1.3 Capability Registry
AC: Consolidate MCP tools and standard API bots into a unified registry for dynamic discovery.

Milestones & Acceptance Criteria

M2: Core Intelligence: Routing & Context (Weeks 6-10)

Core Objective: Implement intent recognition and context-aware multi-turn conversation.

2.1 Contextual Memory Injection
AC: Automatically retrieve the latest N-turn history and inject it into the LLM prompt for context awareness.

2.2 Intent Routing & Fallback
AC: Router correctly dispatches tasks to MCP tools, API bots, or general LLM based on user intent.

2.3 Reference resolution
AC: Resolve pronouns (e.g., "it", "him") and extract missing parameters from session history.

Milestones & Acceptance Criteria

M3: Execution Engine & Hybrid Integration (Weeks 11-13)

Core Objective: Orchestrate external calls across heterogeneous protocols (MCP & API).

3.1 Task Execution (MCP & API)
AC: Execute `call_tool` via MCP and handle standard HTTP requests for vertical domain bots.

3.2 Data Normalization
AC: Transform diverse responses (JSON/Text) into a unified message object (Text/Table/Status).

3.3 Memory Write-back
AC: Automatically update Session DB with execution results to maintain continuous context.

Milestones & Acceptance Criteria

M4: Portal Interface & E2E Validation (Weeks 14-16)

Core Objective: Deliver a seamless UI and ensure end-to-end system reliability.

4.1 Conversational Chat UI
AC: Deploy a clean UI with clear source labeling (e.g., "From: Finance Bot").

4.2 Real-time Feedback
AC: Display active "Calling [Agent]..." status indicators during external execution.

4.3 E2E Integrity & Error Handling
AC: Complete "Input-Route-Execute-Respond" loop with graceful handling of timeouts and API errors.

Future Evolution Plan

1. System Observability & Usage Analytics

- Full-Chain Execution Logging

Log all user queries and tool execution chains for real-time performance monitoring and rapid troubleshooting.

- Product Adoption Analytics

Track usage metrics across different agents to measure user adoption and evaluate the business impact.

2. Developer Enablement & Standardization

- Standardized Onboarding Framework

Establish mandatory API architecture checklists and MCP description guidelines for seamless technical integration.

- Automated Configuration Validation

Implement JSON Schema validation for MCP entries to catch configuration errors before deployment.

Thanks !